

Course Organization and Introduction

Chapter 0 – Hands on Data Analytics for Everyone

August 31, 2026

北师香港浸会大学
BEIJING NORMAL-HONG KONG BAPTIST UNIVERSITY

Welcome to Hands on Data Analytics for Everyone

Course Organization: Instructors for Academic Year 2026/2027



Session	Instructor	Teaching Assistant	Lecture Time (1 hr)	Lab Time (2 hrs)
1001	Dr. Hu, Yujia	Qian Xu	Tue 13:00-13:50, T5-501	Fri 10:00-11:50, V20-101
1002	Dr. Chen, Donald Donglong	Yuwei He	Thu 14:00-14:50, T7-303	Mon 13:00-14:50, T29-101
1003	Dr. Xu, Thomas Canhao	Sinian Luo	Tue 13:00-13:50, T5-405	Fri 10:00-11:50, T8-303
1004	Dr. Li, Zhijian	Lei Hou	Thu 14:00-14:50, T5-405	Mon 13:00-14:50, V20-101
1005	Dr. Xie, Fang	Yingchao Cui	Thu 14:00-14:50, T5-406	Mon 13:00-14:50, T8-303
1006	Dr. Bi, Xiaowen	Yuwei He	Thu 13:00-13:50, T6-505	Tue 8:00-9:50, T29-201

Session 1002: Contact



Instructor: Prof. Donglong Chen

- Office T7-102 / R4
- donglongchen@bnbu.edu.cn
- Office Hours: check timetable here

Teaching Assistant: Miss Yuwei He

Communication Policy:

- **WeChat**: Used for class discussion
- iSpace chat/mail/forum: Used for announcements
- **Email:** Formal communication. Write in English, expect an answer within 24h
- Best use:
  - **Office hours / Tutorials**
  - **Before and after class (5 minutes)**

How to Ask for Questions



- You are **highly encouraged** to come to instructor office hours and TA tutorials for questions
- For short inquiries, WeChat or email me directly donglongchen@bnbu.edu.cn. You can expect a reply within 24h. Email is the official channel of communication, so use a **formal English language**. **Unclear messages will not receive an answer**. Example of email:

Subject: [Hands on Data] Inquiry about …

Email body:

Dear Prof. Chen,

This is Sun Yan from your "Hands on Data" class. I have a question regarding […].

Can you advise?

Regards,
Yan

Tutorials (Assistant Instructor)



Time & Venue

Repeated 2 times a week starting in **week 3**

- To Be Announced
- To Be Announced

Rules / Contents

- Student attendance to the tutorial is **optional**
- Tutorials partially cover the contents of the main lecture / lab (previous week)
- Communication during the tutorials can be in **Chinese or English**
- Students are expected to actively practice and review contents from previous lectures
- Students should ask questions

Course Organization: Grading



Assignments 30%

- 6 assignments, submission on iSpace
- Most assignment are in-class exercises, to be submitted by the end of the class
- Content of assignment is the same or similar to the lab of the same week
- Submission delay: 24h (20% penalty), 48h (40% penalty), >48h (no acceptance)

Participation 10%

The participation consists of

- Class attendance
- Activeness / answer to bonus questions
- Demonstration to lab

Group Project 30%

- Group Size: 3 to 5 members
- Group Work Timing:
  - Form groups (as soon as possible) and do preliminary work by week 8
  - Carry the project work in week 8 – 12 and submit the presentation slides by week 12
  - Present the project in weeks 13-14

Final Exam 30%

- Written examination: General questions in data science, multiple choices and short exercises (e.g., fill missing expressions)
- Exam duration: 2 hours, multiple choice question
- It is **compulsory**

Policies



Late submission

- For assignments and project, **20% penalty for up to 24h** delay, **40% penalty for up to 48h** delay, no late submission will be accepted after 48h
- In case of justified absence (e.g. sickness), inform the course instructor within 24h after deadline and on your return on Campus present approved absence certification (school office). Your instructor **may** decide to void your assignment (instead of assigning a 0 grade) or allow extra time.

Academic honesty

- Copying from others, or allowing others to copy from you, is considered academic plagiarism.

Student conflict / group project :

- The grade for the group project will be assigned individually to each member of the group but within a band
- Group members who do not participate in the group work or fail to present the project will receive 0 grade
- In case of conflict, group members shall try to solve problems by their own before asking the course instructor to mediate

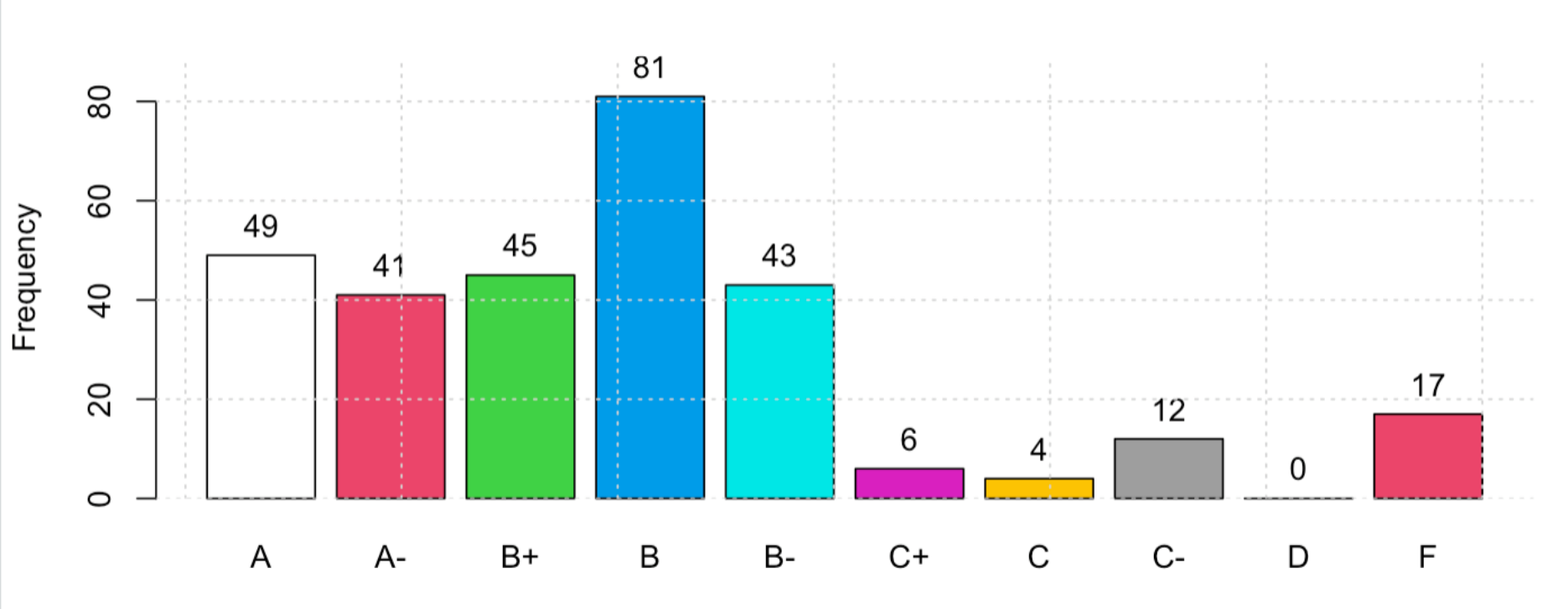
How To Fail This Course

If you want to **fail** this course **do** the following:

- Play with your phone in class (→ 10%)
- Do not come for class (→ 20%)
- Do not do assignments (→ 30%)
- Do not submit your project (→ 50%)
- Do not prepare for exam (→ 70%)
- Copy the exam (→ 99%)



Previous Years Student Performance



Course Plan (Subject to Revision)



Tentative Schedule for GFQR1013 Hands on Data Analytics for Everyone (1002)

Week	Monday	College Note	Mon 13:00-14:00 (T29-101) Lab	Thu 14:00-15:00 (T7-303) Lecture	Makeup Class (Oct. 10th Make-up for Monday's class)	Due Item
1	2026-08-31	Sep 1 (Tue): Courses Start		Introduction + Ch1-Foundations		
2	2026-09-07	Sep 1-14: Add-Drop	ch1-Foundations + Assign 1 (Essay)	ch1-Foundations		Assignment 1 (Essay)
3	2026-09-14	Sep 20 (Sun): Make up Friday	Lab: Knime Intro	ch1-Foundations		
4	2026-09-21	Sep 25 (Fri)-27(Sun): Mid Autumn	Lab: Data Import, filtering	ch2 - Data Processing		
5	2026-09-28	Oct 1(Thu) -7(Wed): National Holiday	Lab + Assign 2 (In-class quiz)			Assignment 2 (In-class quiz)
5	2026-10-05			ch2 - Data Processing	Lab: Knime Data Processing	
6	2026-10-12	Oct 10 (Sat): Make up Monday	Assign 3 (Knime Visualization)	ch2 - Data Processing		Assignment 3 (Knime Visualization)
7	2026-10-19		Lab + Assign 4 (In-class quiz)	ch3 - Modeling		Assignment 4 (In-class quiz)+Project proposal submission
8	2026-10-26	Oct 25-31: Reading Days				
9	2026-11-02		Lab: Linear Regression	ch3 - Modeling		
10	2026-11-09		Assign 5 (Linear Regression)	ch3 - Modeling		Assignment 5
11	2026-11-16		Assign 6 (Classification)	ch3 - Modeling		Assignment 6
12	2026-11-23		Assign 7 (In-class quiz)	Concluding Topics		Assignment 7+ Project presentation submission
13	2026-11-30		Project presentation	Project presentation		
14	2026-12-07	Dec 11 (Fri): Courses End	Project presentation	Review, Exam Preparation		
15	2026-12-14		Exam			
16	2026-12-21					

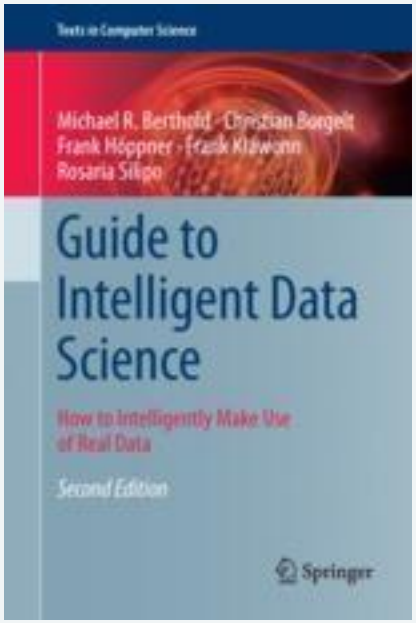
NOTE: This teaching plan is subject to change. Last Updated: **2026/8/31**

References

- Lecture Notes
  - Check iSpace regularly for updates

- Book

  Berthold, M.R., Borgelt, C., Höppner, F., Klawonn, F., Silipo, R. (2020), *"Guide to Intelligent Data Science"*, Springler, **chapters 1-8**



What is Data Analytics?

Which of the Following Make Use of Data Analytics?

1. Autonomous Driving Car



2. Industrial Production



3. Investment in Financial Markets



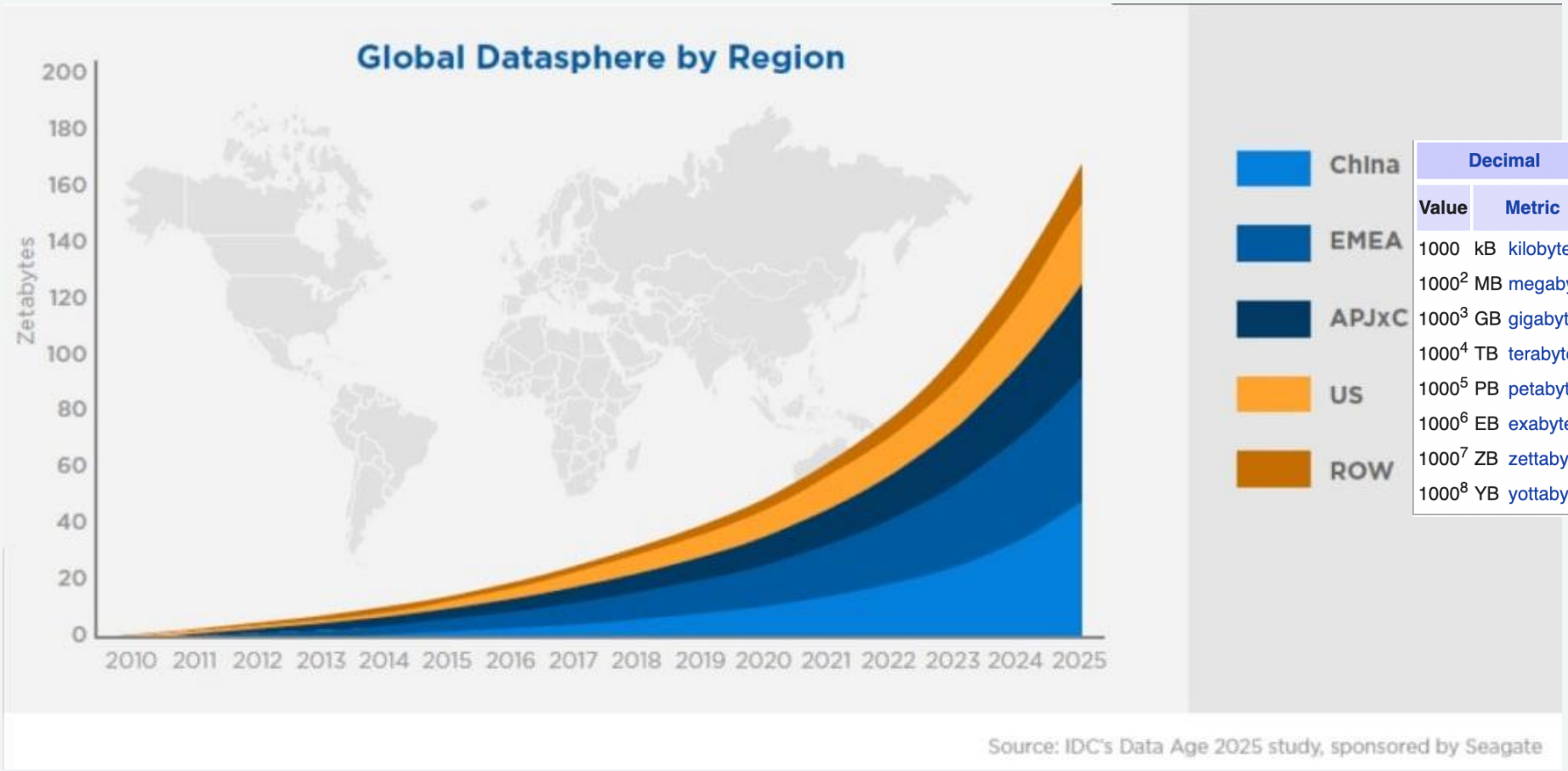
4. Taobao (淘宝)



5. Scientific Research



Data is Everywhere. Can you make use if it?

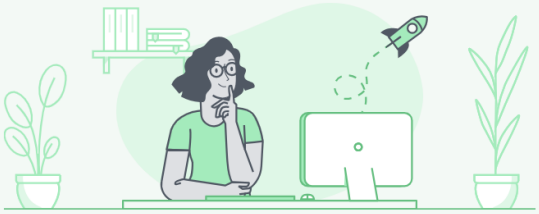


Decimal		
Value		Metric
1000	kB	kilobyte
1000^2	MB	megabyte
1000^3	GB	gigabyte
1000^4	TB	terabyte
1000^5	PB	petabyte
1000^6	EB	exabyte
1000^7	ZB	zettabyte
1000^8	YB	yottabyte

50 Best Jobs in America for 2022

Best Places to Work | Top CEOs | **Best Jobs** | Best Cities for Jobs | Highest Paying Jobs | Share

2022 | United States



Discover Glassdoor's Best Jobs in 2022

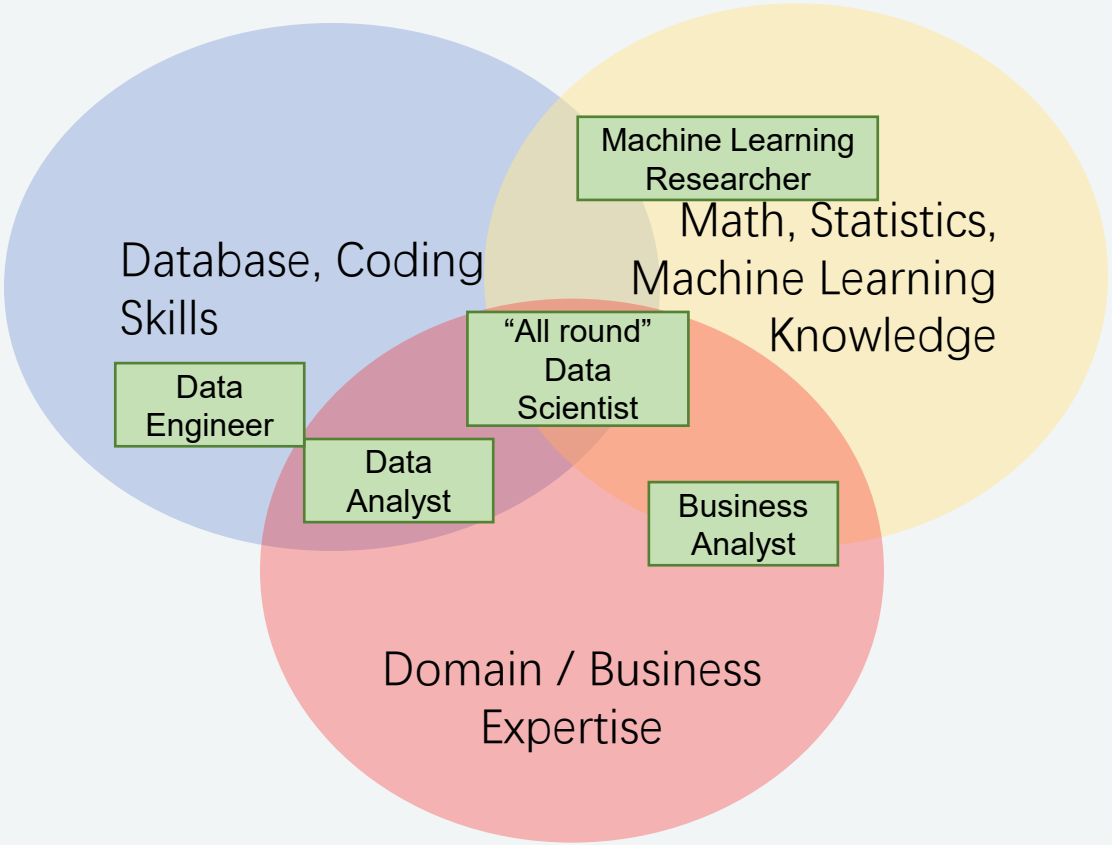
Using Glassdoor's unique data on jobs, salaries, and companies, we compiled a list of the 50 Best Jobs in America to help people find jobs they'll love. Each job stands out for its earning potential (median salary), job satisfaction, and job openings. Are you considering a new position? Check out this comprehensive list to see what jobs made the list this year, and view open jobs at companies across the country.

	Job Title	Median Base Salary	Job Satisfaction	Job Openings	
#1	Enterprise Architect	$144,997	4.1/5	14,021	View Jobs
#2	Full Stack Engineer	$101,794	4.3/5	11,252	View Jobs
#3	Data Scientist	$120,000	4.1/5	10,071	View Jobs
#4	Devops Engineer	$120,095	4.2/5	8,548	View Jobs
#5	Strategy Manager	$140,000	4.2/5	6,977	View Jobs
#6	Machine Learning Engineer	$130,489	4.3/5	6,801	View Jobs
#7	Data Engineer	$113,960	4.0/5	11,821	View Jobs
#8	Software Engineer	$116,638	3.9/5	64,155	View Jobs
#9	Java Developer	$107,099	4.1/5	10,201	View Jobs
#10	Product Manager	$125,317	4.0/5	17,725	View Jobs

Components of Data Science



Data Science Skills



What does a data scientist do?



- Find patterns and trends in datasets to uncover insights
- Create algorithms and data models to forecast outcomes
- Use machine learning techniques to improve the quality of data or product offerings
- Communicate recommendations to other teams and senior staff
- Deploy data tools such as Python, R, or SQL in data analysis